CodeGreedy



OOPs in JAVA



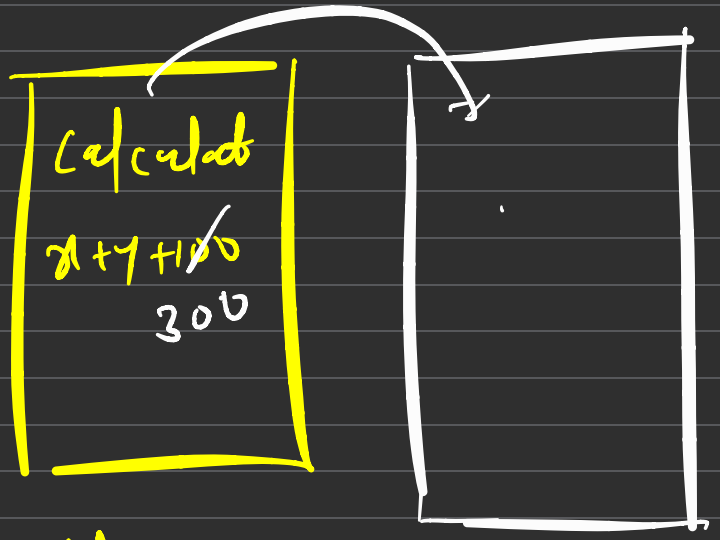
“Success doesn’t come from what you do occasionally, it comes from what you do consistently.”

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C - Procedural oriented

global variable

what



void calculate(x,y)
{
sum = x+y+100;
}

x
10

y
20

=
330

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int x = 10

fun1()

fun2()

main()

60

100

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Class Car

void Car

Car Maruti obj

Car Mercedes

Car Suzuki

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WHY WE NEED OOPS

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Problem with Procedural Programming

✗ 1. No Structure

Everything is just:

Variables

Functions

No clear grouping.

✗ No Reusability

If you want another student:

- You rewrite same code again



No security - anyone can call functions and anyone can use any variable.

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"Procedural programming works for small problems...
but for real-world large systems, OOP is not optional — it's necessary."

Modularity

Break code into small parts (classes)

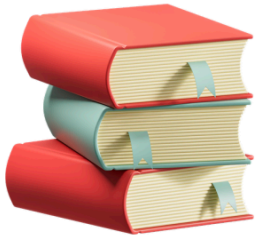
👉 Example:

Student class

Teacher class

Course class

Each is independent.



Scalability

Easy to expand system

Add new feature → just add new class

No need to change entire code

Reusability

Write once, use multiple times

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WHAT IS OOPS

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OOPs is mainly a programming technique which revolves around the real world entities. (classes & object)

Think of it like this:

👉 In real life:

A Car is an object

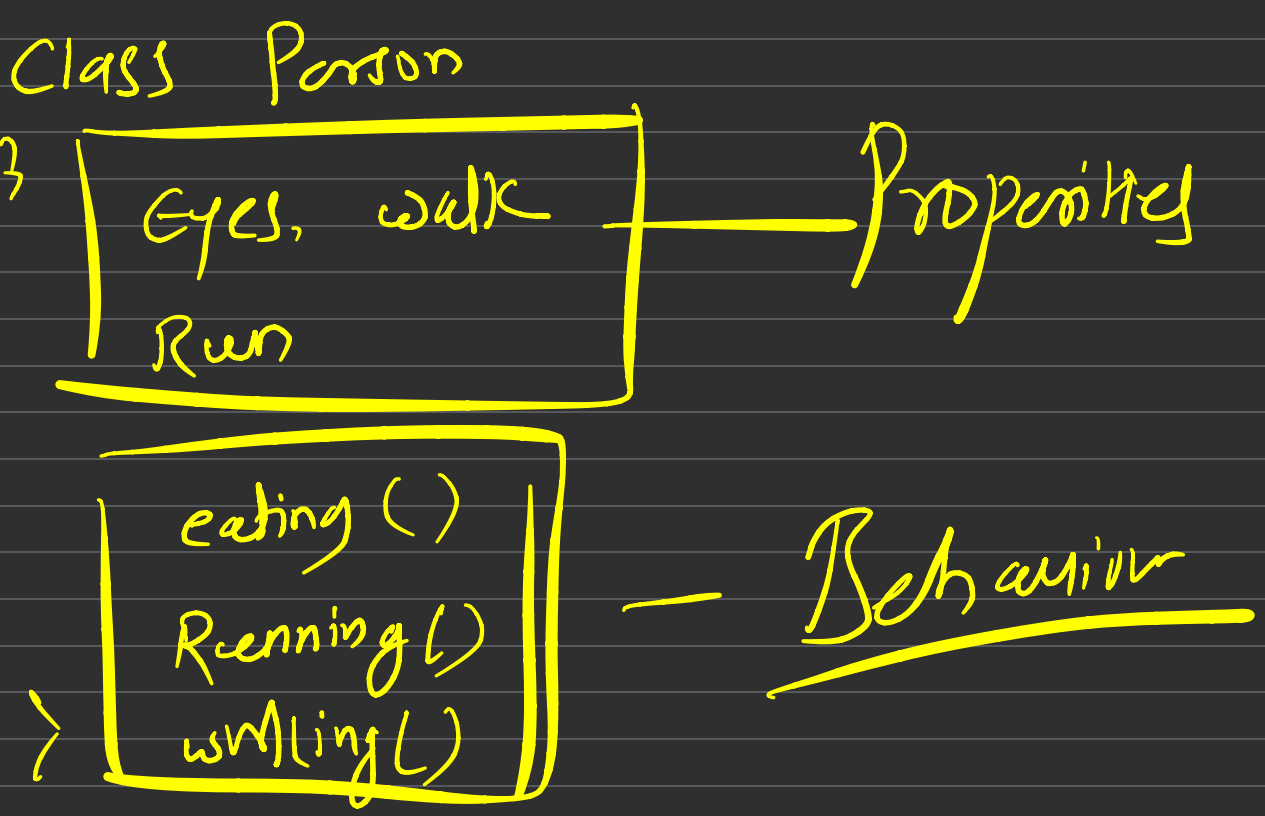
It has properties → color, speed, model

It has behaviors → drive(), brake(), accelerate()



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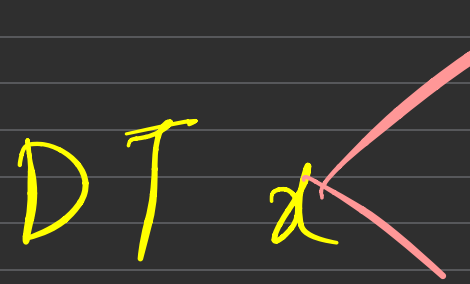


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int x = 10

int[] arr = {8, 9, 8, 1}

User defined

DT x 

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CLASS AND OBJECT

obj

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class

It is user defined data type which mainly contains data members (Variables) and member function together.

methods

It is a logical entity for which no memory is allocated.



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class

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```
class Student {
  // data members
    public string name;
    private int age;

    // member function
    void introduce() {

    }
};
```

Variable

function
method

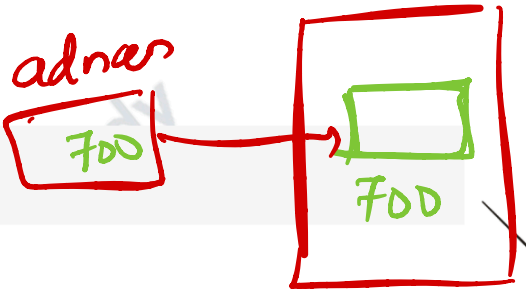


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Object

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It is a instance of the class - which mainly tells what kind of operations we can perform on the class.





Syntax of Object

```
className objectName = new className();
```

Person adnan; = new Person()

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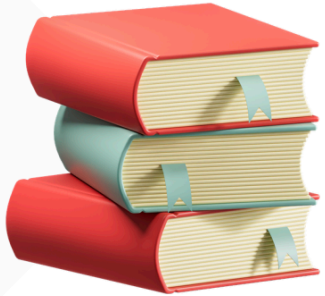
HOW TO ACCESS THE MEMBERS OF CLASS

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To access the members of the class - we mainly use dot (.) operator.

Syntax of access memebers

```
objName.member
```



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int x

Person obj

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Syntax of constrctor

```
class Person{

  int age;
  string name;

  void details(){
    System.out.println("This is a Function");
  }

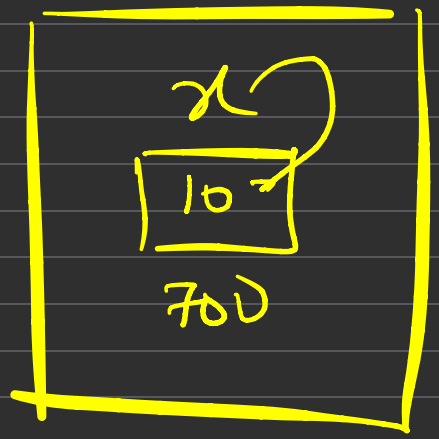
};

Person pramod = new Person();
pramod.details(); // calling a function
```

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int x = 10

print(x) → 10



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ACCESS SPECIFIERS

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Mainly decides the scope of accessebility of members of the class.

Access Specifiers (also called access modifiers) define who can access your class members (variables & methods).



Types of specifiers

default (by default)

private

pubilc

protected

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private

Members will be accessed only inside the class.

public

Members can be accessed inside the class & outside of class.

protected

Members can be accessed inside the class or in derived class.
But it cannot be accessed outside of the class.



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CONSTRUCTOR

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Mainly used to initialize the object of the class.



Should have same name as the class.

Must always be public.

Constructor does not have any return type, not even a void.

Constructor gets called automatically the moment object is created.

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Syntax of constrctor

```
class Person{

  int age;
  string name;

  Person(){
    System.out.println("This is a constructor");
  }

  void details(){
    System.out.println("This is a Function");
  }

};
```



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TYPES OF CONSTRUCTORS

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Default Constructor

Parameterised Constructor

Copy Constructor



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